

L12: Mathematical Analysis of PA Model

Let us now derive mathematically the degree distribution of the Preferential Attachment (PA) model. There are different ways to do these derivations, depending on how rigorous we want to be. Here, we will use the “rate-equation” approach because it is quite general and you can also use it to derive the degree distribution of other growing networks.

Suppose that we add a new node in each time unit, starting from one node at time $t=1$. So, if $N(t)$ represents the number of nodes at time t , we have that $N(t)=t$ for $t>0$.

Let $N_k(t)$ denote the number of nodes of degree k at time t . The probability that a node has degree- k at time t is

$$p_k(t) = \frac{N_k(t)}{N(t)} - \text{the degree distribution changes with time as we add more nodes and edges.}$$

Recall that in the PA model every new node adds m edges to existing nodes. At time t , the total number of edges is $m \times t$ and the sum of all node degrees is twice as large.

The PA model states that the probability $\Pi(k)$ that a new node at time t connects to a specific node v that has degree- k is proportional to k :

$$\Pi(k) = \frac{k}{\sum_j k_j} = \frac{k}{2mt}$$

After we add a new node at time t , the average number of edges that are expected to connect to degree- k nodes is:

$$m \frac{k}{2mt} N(t) p_k(t) = \frac{k}{2} p_k(t)$$

because a new node brings m new edges, and the average number of nodes of degree- k is $N(t) p_k(t)$.

This is also the average number of nodes of degree- k that get a new edge and become nodes of degree- $(k+1)$ -- assuming that each node gets at most one new edge.

Similarly, some nodes of degree- $(k-1)$ will get a new edge and they will become nodes of degree- k .

Using the previous expression again, the expected number of such nodes is $\frac{k-1}{2} p_{k-1}(t)$.

So, considering how many nodes of degree- k we have at time t , how many nodes of degree- $(k-1)$ become nodes of degree- k at time $t+1$, and how many nodes of degree- k become nodes of degree- $(k+1)$ at time $t+1$, we can write that the expected number of degree- k nodes at time $t+1$ as:

$$(N+1) p_k(t+1) = N p_k(t) + \frac{k-1}{2} p_{k-1}(t) - \frac{k}{2} p_k(t)$$

The previous expression applies to all degrees $k>m$ -- but it cannot be used for the minimum possible degree m because there are no nodes with degree $m-1$ (remember that even a newly born node has m edges).

Instead of having nodes of degree- $(m-1)$ that are “promoted” to nodes of degree- m , we add exactly ONE node of degree- m at each time step.

So the expected number of degree- m nodes at time $t+1$ is:

$$(N+1) p_m(t+1) = N p_m(t) + 1 - \frac{m}{2} p_m(t)$$

Note that the previous two expressions give us a recursive process for computing $p_k(t)$ for any value of $k \geq m$ and for any time $t > 0$.

What happens asymptotically, as the network size N increases? It can be shown that the probability distribution $p_k(t)$ becomes stationary, meaning that it does not change with time (we will not prove this step however). We will also see some numerical results that support this claim shortly.

So, instead of $p_k(t)$ we can write that $\lim_{t \rightarrow \infty} p_k(t) = p_k$.

The expression for nodes of degree- m becomes asymptotically:

$$(N+1) p_m(t+1) - N p_m(t) \rightarrow p_m = 1 - \frac{m}{2} p_m$$

which is equivalent to:

$$p_m = \frac{2}{m+2}$$

And the expression for nodes of degree- k with $k > m$ becomes asymptotically:

$$(N+1)p_k(t+1) - Np_k(t) \rightarrow p_k = \frac{k-1}{2}p_{k-1} - \frac{k}{2}p_k$$

which is equivalent to:

$$p_k = \frac{k-1}{k+2}p_{k-1}, \quad \text{for } k > m$$

or

$$p_{k+1} = \frac{k}{k+3}p_k, \quad \text{for } k \geq m$$

when $k \rightarrow k+1$.

We now have a recursive formula that we can easily solve using induction to show that the probability of degree- k nodes is:

$$p_k = \frac{2m(m+1)}{k(k+1)(k+2)} \quad \text{for } k \geq m \quad (1)$$

This is the degree distribution equation for the PA model, at least for large networks.

Note that for large degrees (large k), this expression becomes a power-law with exponent 3, i.e., $p_k \approx c k^{-3}$, where $c = 2m(m+1)$. **This is the main result of these derivations.**

The PA model generates power-law networks – but with a fixed exponent. Further, this exponent is equal to 3, which means that the first moment (mean) of the degree distribution is finite – but any higher moment (including the variance) diverges.

Additionally, note that the degree exponent does not depend on the parameter m . That parameter only controls the minimum degree of the distribution.

Let us now look at some numerical results to get a better insight in the previous results.

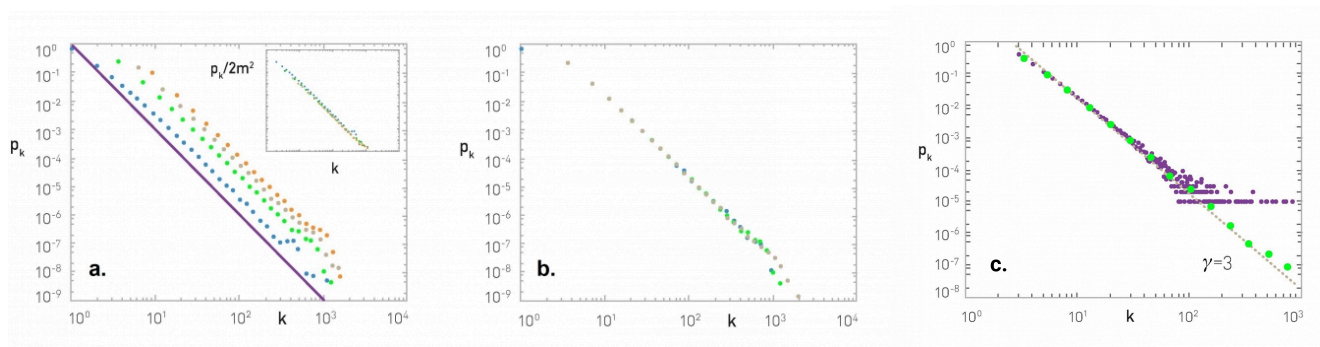


Figure (a) shows what happens as we vary the value of m (blue: $m=1$, green: $m=3$, grey: $m=5$, orange: $m=7$). The distributions are parallel to each other, having the same exponent. The inset plot shows what happens if we plot $\frac{p_k}{2m^2}$ – the effect of m disappears, meaning that p_k is proportional to $2m^2$, as also predicted by our earlier mathematical derivation.

Figure (b) shows what happens when we vary the size of the network N (blue: $N=50,000$, green: $N=100,000$, grey: $N=200,000$) – all of the plots have $m=3$. The resulting degree distributions are practically indistinguishable, supporting our earlier claim that the degree distribution becomes stationary (independent of time) at least for large networks.

Figure (c) shows the degree distribution of the PA network with $N=100,000$ nodes and $m=3$. The purple dots are the linearly-binned plot of the empirical degree distribution, while the green dots represent the log-binned version of the same plot. Note that the latter shows more clearly that the degree distribution behaves as a power-law with exponent 3.

Food For Thought

a) Solve the recursive equations given above to show Equation (1).

b) Use the same analytical approach to derive the degree distribution for a version of the PA model that applies to **directed networks**, as follows:

The probability that a new node connects with a directed edge to a specific node of in-degree $k_{in}(i)$ is

$$\Pi[k_{in}(i)] = \frac{k_{in}(i)+A}{\sum_j [k_{in}(j)+A]}$$

where A is the same constant for all nodes. Each new node brings m directed links.